



GOA UNIVERSITY

B.E - (Mechanical) (Sem-VII)(Revised Course 2019-2020)**Paper Title: OPEN ELECTIVE Additive Manufacturing****Paper Code: OME725****[Duration: 3 Hours]****[Total Marks: 100]**

- Instructions: 1) Answer any two questions each from **Part A** and **Part B** and one question from **Part C**
 2) Assume missing data, if any
 3) illustrate your answers with neat sketches wherever required.

PART AAnswer any **TWO** questions from the following:

- Q1 (A) Discuss the significance of additive manufacturing in modern product development. What are the key advantages it offers over traditional manufacturing methods? **07**
- (B) A medical device start-up needs to manufacture highly detailed surgical guides with complex geometries for orthopaedic surgeries. Which liquid-based additive manufacturing system would be most suitable, and why? **07**
- (C) Compare and contrast additive manufacturing with subtractive manufacturing. What are the strengths and weaknesses of each approach in terms of cost, material usage, and design flexibility? **06**
- Q2 (A) Explain the different classifications of additive manufacturing technologies. How do these classifications impact the choice of process for specific applications? **07**
- (B) An automotive company plans to manufacture aerodynamic test models for wind tunnel testing using ABS filament with a 1.75 mm diameter. Suggest a suitable process along with a neat sketch, describe the working principle, and explain the advantages and disadvantages associated with this process. **07**
- (C) A medical device company needs to produce patient-specific, biocompatible metal implants for providing strength to the bones. Which powder-based additive manufacturing process should be chosen to ensure high accuracy, strength and with higher density. **06**
- Q3 (A) Explore the various applications of additive manufacturing across different industries. Provide examples of how this technology is transforming product design and manufacturing. **07**

- (B) A defense manufacturer needs to repair a damaged high-strength metal part in a cost-effective manner. Which powder-based additive manufacturing method would be most suitable for this application? **07**
- (C) A company specializing in electronics enclosures needs to produce detailed plastic casings with high dimensional accuracy. How would the Solid Ground Curing (SGC) process ensure that the required precision is achieved? **06**

PART B

Answer any **TWO** questions from the following:

- Q4 (A) An electronics company needs to rapidly prototype detailed circuit board casings with smooth surface finishes. How would a thermal jet printer-based 3D modeler be suitable for this application, and what are its limitations? **07**
- (B) A company needs to produce a small batch of customized injection molds quickly. Which indirect rapid tooling method would be most cost-effective and why? **07**
- (C) A manufacturer is looking to digitize their existing physical parts for reverse engineering. What digitization technique would be most suitable for capturing detailed geometric data? **06**
- Q5 (A) A manufacturer specializing in dental prosthetics needs to produce highly accurate wax patterns. How would Sanders Model Market 3-D Printers fulfil this need, and what advantage does it provide over other modellers? **07**
- (B) A company requires soft tooling to prototype injection-moulded parts quickly. but with some flexibility for design iterations. What advantages does soft tooling offer compare to hard tooling in this scenario? **07**
- (C) In preparing a CAD model for additive manufacturing, what is the significance of part orientation, support generation, and how can it affect the final product? **06**
- Q6 (A) A manufacturer notices significant dimensional inaccuracies in parts built using FDM additive manufacturing. List down the process parameters which contributes to this inaccuracies and How it can help to reduce errors in the final parts? **07**
- (B) In the field of computer-aided tissue engineering (CATE), how does bio-additive manufacturing contribute to the development of functional tissue constructs? **07**
- (C) A designer is preparing a complex 3D model for 3D printing and needs to save it in the correct format. What are the key characteristics of STL files that make them suitable for additive manufacturing? **06**

PART C**(Answer any ONE question from the following)**

- Q7 (A) Compare the suitability of Stereo lithography (SLA) and Fused Deposition Modelling (FDM) for producing functional prototypes. What factors should be considered when choosing between the two methods? **07**
- (B) A civil engineering company is looking for an additive manufacturing process which can build prototypes from the paper. Suggest the process with the aid of a neat sketch, working principle, advantages & disadvantages. **07**
- (C) A precision tool manufacturer is evaluating the use of Object Quadra Systems for creating small, highly detailed prototypes. What features of this system make it suitable for such applications? **06**
- Q8 (A) A researcher is studying case studies in bio-additive manufacturing to understand the challenges faced in creating customized prostheses. What common challenges might they encounter in the design and production phases? **07**
- (B) A company is considering options for producing a limited run of custom parts. What is one primary advantage of using soft tooling over hard tooling in this scenario? **07**
- (C) Write a note on **06**
- Rapid manufacturing process optimization
 - Software's for RP